

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 14-01 NRC ESG-2

SCENARIO NUMBER: 14-01 NRC ESG-2

EFFECTIVE DATE: See Approval Dates

EXPECTED DURATION: 65 minutes

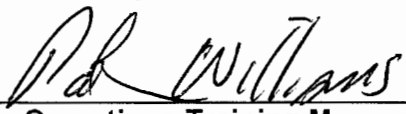
REVISION NUMBER: 00

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

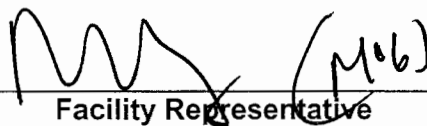
Revision Summary
New Issue for 14-01 ILOT NRC exam

PREPARED BY: G Gauding
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9-2-15
Date

APPROVED BY: 
Operations Training Manager

10-7-15
Date

APPROVED BY:  (Meb)
Facility Representative

10-9-15
Date

SCAN OF SIGNED SCENARIO COVER SHEET

I. OBJECTIVES

- A. Given the order, OPERATE auxiliary building ventilation in accordance with the approved station procedures.
- B. Given the order or indications of a pressurizer control system malfunction, perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S1/S2.OP-AB.PZR-0001.
- C. Given indication of a pressurizer control system malfunction, DIRECT the response to the malfunction in accordance with S1/S2.OP-AB.PZR-0001.
- D. Given the order or indications of a fire, as the nuclear control operator PERFORM the control room actions in accordance with S1/S2.OP-AB.FIRE-0001.
- E. Given the order or indications of a fire, DIRECT the control room actions in accordance with S1/S2.OP-AB.FIRE-0001.
- F. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with the approved station procedures.
- G. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with the approved station procedures.
- H. Given the order or indications of a steam generator tube rupture (SGTR), perform actions as the nuclear control operator to RESPOND to the tube rupture in accordance with the approved station procedures.
- I. Given indication of a steam generator tube rupture (SGTR), DIRECT the response to the SGTR in accordance with the approved station procedures.

II. MAJOR EVENTS

- A. 21 ABV Exhaust Fan stopped.
- B. Controlling PZR Pressure Channel fails low.
- C. Main Generator Radio Frequency high.
- D. Fire Protection Actuation (deluge) at Main Power Transformer
- E. Inadvertent Safety Injection.
- F. SGTR after SI termination.

III. SCENARIO SUMMARY

- A. The crew takes the watch with the unit at 100%, MOL. 26 SW pump is C/T for strainer cleaning. Solar Flares are occurring. SMD K-5 is in effect for the next 4 hours.
- B. Shortly after assuming the watch, 21 Aux Building Exhaust fan stops. The crew will refer to the ARP for Air Flow Low for 21 ABV Exhaust fan, and place an alternate ABV Exhaust fan in service. The CRS will identify the ABV system LCO 3.7.7.
- C. Once the ABV system malfunction has been addressed, the controlling PZR Pressure channel fails low. The crew will identify the failure and the CRS will enter S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction. The crew will take manual control of PZR pressure, restore PZR to normal band, swap to an operable channel and restore PZR control to auto. The CRS will identify the requirement to close 2PR1 PZR PORV Block valve, and remove power from the Block valve. The crew will initiate actions to remove the failed channel from service. The CRS will identify LCOs 3.3.1.1 for Rx Trip Instrumentation, 3.3.2.1 for ESFAS Instrumentation, and 3.4.5 for the inoperable 2PR1.
- D. After addressing the PZR Pressure channel failure, the control room will receive Aux Annunciator point 0901, Gen Radio Freq High. The CRS will enter S2.OP-AB.GEN-0001, Main Generator Abnormal Stator Conditions. The crew will determine Main Generator thermocouple temperatures are elevated and initiate a load reduction to lower temperatures. After the downpower is commenced, a Main Power Transformer (MPT) Sudden Pressure alarm will occur, and a deluge actuation at the MPT occurs. The CRS will order a Rx trip with power >P-9.
- E. The crew will perform immediate actions of 2-EOP-TIP-1, Reactor Trip or Safety Injection, and transition to 2-EOP-TRIP-2 with no SI required. A second channel of PZR pressure fails low, causing an unwarranted Safety Injection on Low PZR pressure. The crew will return to EOP-TRIP-1.
- F. The crew will perform TRIP-1 plant stabilization and diagnostics, and transition to 2-EOP-TRIP-3, Safety Injection Termination. The crew will terminate the Inadvertent SI within the Critical Action Time (CT-1), and re-establish letdown within the Critical Action Time.
- G. After re-establishing letdown, a SGTR will occur, and the crew will re-start ECCS pumps as required.
- H. The scenario will be terminated after the crew has restarted ECCS pumps.

IV. INITIAL CONDITIONS

___ IC-232

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
___ 1	VC1and VC4 C/T
___ 2	RCPs (SELF CHECK)
___ 3	RTBs (SELF CHECK)
___ 4	MS167s (SELF CHECK)
___ 5	500 KV SWYD (SELF CHECK)
___ 6	SGFP Trip (SELF CHECK)
___ 7	23 CV PP (SELF CHECK)
___ 8	26 SW pump C/T.
___ 9	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
	1	EVENT ACTION: MONP254 <10 // CONT ROD BANK C < 10 (RX TRIP) COMMAND: PURPOSE: <update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
01	PR0016A PZR PRESS CH I (PT455) FAILS H/L	N/A	N/A	N/A	RT-3	1700
02	AN3901 AAS 901 FAILS - :GENERATOR RADIO FREQUENCY HIGH	N/A	N/A	N/A	RT-5	AAS POINT FAILS/OVRD TO ON
03	AN0529 SER 529 FAILS - :H15 MAIN POWER TRANSFORMER PHASE 1 SUDDEN PRESSURE	N/A	N/A	N/A	RT-7	SER POINT FAILS/OVRD TO ON
04	AN0316 SER 316 FAILS - :A15 FIRE PUMP 1 AND OR 2 RUN	00:03:15	N/A	N/A	RT-7	SER POINT FAILS/OVRD TO ON
05	AF0353C 23 AFP FAILURE TO AUTO START ON ANY (ALL) SIGNALS	N/A	N/A	N/A	N/A	
06	AF0182A 21 AFP PRESS OVRD PROT FAILS	N/A	N/A	N/A	N/A	
07	PR0016B PZR PRESS CH III(PT457)FAILS H/L	N/A	N/A	N/A	RT-12	1700
08	SG0078D 24 STEAM GENERATOR TUBE RUPTURE	N/A	N/A	N/A	RT-14	1200
09	RP318G3 21 Aux Bldg Exhaust Fan Fails to Start on SEC	N/A	N/A	N/A	N/A	
10	RP0060A FAILURE OF TRAIN "A" RX TRIP BREAKER TO TRIP	N/A	N/A	N/A	N/A	
11	AF0353B 22 AFP FAILURE TO AUTO START ON ANY (ALL) SIGNALS	N/A	N/A	N/A	N/A	

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
01	FP07D FIRE - MAIN TRANSFORMERS (46)	00:03:00	N/A	N/A	RT-7	FIRE
02	SW43D 26 SW PUMP BKR CONTROL POWER	N/A	N/A	N/A	N/A	OFF
03	PR34D PORV STOP VALVE 2PR6 TAGGED	N/A	N/A	N/A	RT-10	TAGGED
04	SP01A1 Prevailing Wind Speed	N/A	N/A	N/A	N/A	10
05	SP01A2 Prevailing Wind Direction	N/A	N/A	N/A	N/A	160

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
_____ 01	AG03 F DI 21 AUX BLDG EXHAUST FAN STOP	N/A	N/A	N/A	RT-1	ON

OTHER CONDITIONS:

Description

_____ 1.

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Loss of 21 ABV Exhaust Fan			
Simulator Operator: Insert RT-1 on direction from Lead Evaluator.			
O/R AG03 OVDI 21 ABV Exh Fan Stop PB Final Value: ON			
	RO reports unexpected AIR FLOW LO alarm for 21 ABV Exhaust Fan.		
	CRS dispatches an operator to check status of breaker 2AX1AX12X.		
Role Play: 2 minutes after being dispatched, report breaker 2AX1AX12X is tripped open.			
	CRS refers to OP-SA-108-106-1001, Large Motor Starting Criteria and Protective Circuit Breaker Reset and Reclosure Policy, and determines the cause of the breaker trip must be investigated prior to reclosure. IF the CRS directs reset and closing of 21 ABV Exhaust Fan breaker, it will not shut.		
Note: OP-SA-108-106-1001, page 3, Section 4.2 contains the discussion about breaker reclosure. LCO 3.7.7 has a 14 day action time.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Continue to next event after LCO has been identified. 2. Controlling PZR Pressure Channel Fails Low	Crew places the alternate ABV Exhaust Fan in service IAW S2.OP-SO.ABV-0001, Auxiliary Building Ventilation System Operation.		
	RO starts 23 ABV Exhaust Fan.		
	CRS enters LCO 3.7.7 for less than the full complement of Aux Building Ventilation fans.		
Simulator Operator: Insert RT-3 on direction from Lead Evaluator. MALF: PR0016A PZR Press CH I (PT-455) fails H/L Severity: 1700			
	RO reports unexpected OHAs D-16 RC PRESS LO, E-12 PZR PRESS LO, and E-28 PZR HTR ON PRESS LOW.		
	RO identifies failure of controlling PZR Pressure channel.		
	RO requests permission from CRS prior to placing the Master Pressure Controller in MANUAL to mitigate the transient		
	CRS enters S2.OP-AB.PZR-0001, Pressurizer		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	Pressure Malfunction.		
	CRS directs initiation of Attachment 1 CAS.		
	RO reports POPS not in service.		
	RO reports controlling PZR Pressure Channel I failed Low.		
	RO places Master Pressure Controller in MANUAL if not previously performed and adjusts demand per Attachment 2.		
	RO selects PZR Pressure Channel III for control.		
	RO restores the Master Controller to AUTO.		
	CRS enters TSAS(s) 3.3.1.1 Action 6, 3.3.2.1.b Action 19 and 3.4.5 Action b.		
	CRS directs removing failed channel from service.		
	RO verifies alternate control channel selected for control.		
	RO selects channel other than channel I for PZR Pressure recorder.		
	RO selects channel other than channel I for Differential Temperatures recorder.		
	RO reports bistables must be tripped to complete procedure.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Continue to next event after 2PR6 is closed, or at Lead Evaluator's discretion.			
	CRS directs RO to close 2PR6.		
	CRS directs WCC to de-energize 2PR6.		
Simulator Operator: 3 minutes after being directed, enter RT-10 then report 2PR6 de-energized. REMOTE: PR34D = TAGGED			
3. Main Generator Radio Frequency High			
Simulator Operator: Insert RT-5 on direction from Lead Evaluator. MALF: AN3901, AAS 901 FAILS: Generator Radio Frequency High Severity: 2			
	RO reports unexpected OHA A-41.		
	PO reports alarm is Point 0901, Gen Radio Freq High.		
	CRS enters S2.OP-AB.GEN-0001, Main Generator Abnormal Stator Conditions.		
	CRS directs initiation of Attachment 1 CAS.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO checks Stator T/C and RTD temperatures on SPDS.		
Floor Evaluator: When PO checks SPDS, verify they can properly access temperature displays, then provide pages 9-14 of S2.OP-PT.GEN-0003, labeled "Copy 1", which indicates 3 sequential temperature above assigned limit but less than <10 above assigned limit.			
	PO reports the following condition listed in step 3.2 is present: - 3 sequential stator T/Cs exceed assigned temperature limits by <10°F		
	Crew may dispatch operator to report local reading of RF Monitor.		
Role Play: If dispatched, then 2 minutes later report the following: Local Radio Frequency Monitor is reading just above 50%, and the last log reading was 5% taken 13 hours ago. The reading appears to have started rising 1 hour ago and the chart recorder indicates it is still rising slowly. The Red and Yellow lights are lit and there was an audible alarm until you acknowledged it.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
AB.LOAD steps	CRS determines a load reduction is necessary to lower Radio Frequency Monitor to <50%.		
	CRS briefs load reduction to be performed using S2.OP-AB.LOAD-0001, Rapid Load Reduction, or S2.OP-IO.ZZ-0004, Power Operation.		
	CRS enters S2.OP-AB.LOAD-001, Rapid Load Reduction.		
	CRS directs initiation of Attachment 1 CAS.		
	CRS directs RO and PO to initiate and coordinate load reduction.		
	RO calculates boron addition required for load reduction.		
	RO commences RCS boration.		
	PO initiates turbine load reduction to the target load at the rate specified by CRS.		
	RO maintains Tavg on program using control rods and boration.		
	RO energizes PZR heaters.		
Role Play: After power reduction is started, if Field Operator is asked about current radio frequency magnitude, give indications of			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
stable or slowly lowering radio frequency.			
4. Main Power Transformer Phase 1 Sudden Pressure / Fire Protection Deluge actuation			
Simulator Operator: Insert RT-7 on direction from Lead Evaluator. MALF: AN0529 SER 529 FAILS - :H15 MAIN POWER TRANSFORMER PHASE 1 SUDDEN PRESSURE REMOTE: FP07D FIRE - MAIN TRANSFORMERS (46) Delay: 00:3:00 MALF: SER 316 FAILS - :A15 FIRE PUMP 1 AND OR 2 RUN Delay: 00:03:15			
	PO reports unexpected OHA H-15, MPT Ø1 TRBL, and refers to Alarm Response.		
	RO reports CRT point 529 MPT Phase 1 Sudden Pressure is in alarm.		
Note: One of the potential automatic actions listed in ARP for sudden pressure is FP Deluge actuation.			
	CRS dispatches an operator to investigate alarm.		
	RO reports unexpected OHA A-7 FIRE PROT		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: OHA A-15 alarms when either of the Diesel Drive Fire Pumps starts. Role Play: Shortly after the Fire Pump ½ Run OHA A-15 annunciates, call control room and report there is water spraying all over MPT Phase 1, there is heavy arcing and sparking, and you have left the area for safety concerns.	FIRE, with coded audible alarm indicated on 2RP5 Fire Protection Panel.		
	PO reports Zone 46 for MPT is affected area.		
	RO reports unexpected OHA A-15 FIRE PUMP 1/2 RUN.		
	CRS determines the Main Generator must be removed from service, and power is >P-9.		
	CRS directs the RO to trip the reactor.		
	RO trips the reactor.		
	RO performs Immediate Actions of TRIP-1 - Confirms the Rx trip - Trips the Main Turbine - Reports at least one 4KV Vital bus energized. - Reports SI is not actuated, and reports SI is not required.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
5. No AFW flow post trip	CRS and RO verify performance of immediate actions.		
	CRS transitions to EOP-TRIP-2, Reactor Trip Response.		
	PO reports 21 AFW pump running with no discharge flow, and 22 and 23 AFW pumps have not started.		
	PO starts at least one of 22 and 23 AFW pump to establish at least 22E4 lbm/hr AFW flow and maintains at least 22E4 lbm/AFW flow until at least one SG NR level is >9%, then maintains intact SG NR level 9-33%.		
	PO stops/ensures stopped 21 and 22 SGFP.		
Role Play: Prior to inserting RT-12, call as Fire Protection Operator to report there is no fire currently at Unit 2 MPT area. 6. Inadvertent SI	PO may request to depress Pressure override Defeat for 21 AFW pump, and if done, will establish AFW flow from 21 AFW pump.		
	RO reports RCP status and RCS temperature status.		
Simulator Operator: Insert RT-12 on direction from Lead Evaluator.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
MALF: PR0016B PZR PRESS CH III(PT457)FAILS H/L Severity: 1700			
Evaluators mark time SI is actuated to track Critical Task performance. _____:_____:_____			
Note: Actions have already been taken for the ABV Exhaust fan.	RO reports alarms and indications that an automatic Safety Injection has occurred, and backs up SI.		
	CRS goes to EOP-TRIP-1 from TRIP-2 CAS of SI initiation.		
	CRS and RO verify performance of immediate actions.		
	PO reports SEC loading not complete, but all available equipment started.		
	PO reports AFW flow status and maintains 22E4 lbm/hr until at least one SG NR level is 9%, then maintains 9-33%.		
	PO reports valve groups in Table B are in safeguards position.		
	RO reports 21 and 22CA330 are shut.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: RCS pressure will rise and be controlled by PZR PORV 2PR2 cycling in auto.	RO reports containment pressure has remained less than 15 psig.		
	RO reports no indication of high steamline flow.		
	PO reports all 4KV Vital buses energized.		
	RO reports CAV in AP mode.		
	RO runs two switchgear room supply and one exhaust fans.		
	RO reports two or more CCW pumps running.		
	RO reports RHR is not aligned for cold leg recirc.		
	RO reports RCS pressure is >1540 psig.		
	PO reports AFW flow status and maintains 22E4 lbm/hr until at least one SG NR level is 9%, then maintains 9-33%.		
	RO reports RCP in service.		
	RO reports RCS temp stable at or trending to 547°F.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Simulator Operator: 2 minutes after being dispatched, delete MALF RP0060A and ensure RTB A opens.	RO reports RTB "A" has not opened.		
	CRS dispatches an operator to locally open RTB "A".		
	RO reports PZR PORV operating correctly based on current RCS pressure, and reports Block Valves opens.		
	RO reports RCPs in service.		
	RO reports both PZR Spray Valves shut.		
	RO reports RCS pressure is > 1350 psig.		
	PO reports no indications of a faulted SGs.		
	RO reports no indication of ruptured SGs.		
	RO reports no rad monitors in Table F in warning or alarm.		
	RO reports containment pressure is < 4 psig.		
	RO reports containment sump level is < 46%.		
	RO reports subcooling greater than 0°F.		
	PO reports AFW flow is at least 22 E4 lbm/hr		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Channels I and III PZR pressure are failed low, so there will be no demand for spray flow.	or at least one SG NR level is >9%.		
	RO reports RCS pressure is stable or rising, and PZR level is >11%.		
	CRS transitions to EOP-TRIP-3, SI Termination.		
	RO resets SI and Phase A isolation.		
	RO reports Phase B isolation reset, and opens 21 and 22CA330's.		
	RO reports no automatic control for PZR Pressure, and manually controls RCS pressure.		
	RO reports PZR spray valves not failed open.		
	PO resets each SEC, and reports 230V control centers reset.		
	PO stops all but 21 or 22 Charging pump.		
CT#1 (Salem Critical Operator Action) Terminate an Inadvertent SI by stopping all but 21 or 22 charging pump within 23 minutes of SI initiation.			

Evaluator/Instructor Activity		Expected Plant/Student Response	SBT LOG	Comment
SAT _____ UNSAT _____				
		RO reports RCS pressure stable or rising.		
		RO reports charging pump suction is aligned to the RWST.		
		RO ensures open 2CV139 and 2CV140 charging pump mini flow valves.		
		RO shuts BIT isolation valves 2SJ4, 2SJ5, 2SJ12, and 2SJ13.		
		RO shuts 2CV55 Charging FCV.		
		RO opens 2CV68 and 2CV69 Charging Discharge valves.		
		RO adjusts 2CV55 to maintain PZR level >25%.		
		RO reports PZR level is stable or rising.		
		RO reports RCS pressure >1540 psig and stable or rising.		
		RO stops both SI and both RHR pumps.		
		RO reports RCS subcooling > 0 °F and PZR level <11%.		
		RO reports no CS pumps running.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports PZR level >25%.		
	RO opens 2CV2 and 2CV277 and places in auto.		
	RO ensures open 2CV7.		
	RO raises charging flow to at least 87 gpm.		
	RO places one letdown orifice in service while maintaining letdown pressure at 300 psig, places 2CV18 letdown pressure control valve in auto, and reports normal letdown is in service.		
CT#2 (Salem Critical Operator Action) Terminate an Inadvertent SI by re-establishing Letdown within 45 minutes of the inadvertent SI. _____:_____:_____ SAT _____ UNSAT _____			
7. SGTR initiation	RO ensures CVCS auto makeup in Auto.		
Simulator Operator: Insert RT-14 on direction from Lead Evaluator: MALF: SG0078D 24 STEAM GENERATOR TUBE RUPTURE			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: The Basis Document for TRIP-3 does not specify which or how many ECCS pumps to start. The CRS should base their decision on which, if any, ECCS pumps to start, on RCS pressure and trend. For instance if RCS pressure is >1540 psig, the only ECCS pump which would cause an effect on PZR level would be a centrifugal charging pump which is O/S. Terminate the scenario when the transition to SGTR-1 is made.	RO reports lowering RCS pressure and PZR level (if on scale).		
	PO reports rising level in 24 SG.		
	CRS directs CAS action to start ECCS pumps as necessary and go to EOP-SGTR-1.		

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. OP-SA-108-106-1001, Large Motor Starting Criteria and Protective Circuit Breaker Reset and Reclosure Policy
- F. S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction
- G. S2.OP-AB.GEN-0001, Main Generator Stator Abnormality
- H. S2.OP-PT.GEN-0003, Main Generator Stator Temperatures.
- I. 2-EOP-TRIP-1, Reactor Trip or Safety Injection
- J. 2-EOP-TRIP-2, Reactor Trip Response
- K. 2-EOP-TRIP-3, SI Termination

**ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY**

MODE: 1 POWER: 100 RCS BORON: 864 MWe 1220

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

None

ABNORMAL PLANT CONFIGURATIONS:

Solar Flares are occurring. SMD K-5 in effect for next 4 hours.

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

No penalty minutes in the last 24 hrs.

PRIMARY:

None

SECONDARY:

None

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

26 SW pump C/T for strainer cleaning.

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
___ With Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.
- ___ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.
- ___ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3**CRITICAL TASK METHODOLOGY**

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 14-01 ILOT NRC ESG-2 REVIEWER: B Blose

Initials Qualitative Attributes

- | | | |
|----|-----|---|
| BB | 1. | The scenario has clearly stated objectives in the scenario. |
| BB | 2. | The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events. |
| BB | 3. | The scenario consists mostly of related events. |
| BB | 4. | Each event description consists of: <ul style="list-style-type: none"> • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point |
| BB | 5. | No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event. |
| BB | 6. | The events are valid with regard to physics and thermodynamics. |
| BB | 7. | Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives. |
| BB | 8. | The simulator modeling is not altered. |
| BB | 9. | All crew competencies can be evaluated. |
| BB | 10. | The scenario has been validated. |
| BB | 11. | If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario. |
| BB | 12. | ESG-PSA Evaluation Form is completed for the scenario at the applicable facility. |

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

Note: The quantitative attribute target ranges that are specified on the form are not absolute limitations; some scenarios may be an excellent evaluation tool, but may not fit within the ranges. A scenario that does not fit into these ranges shall be evaluated to ensure that the level of difficulty is appropriate. (ES-301 Section D.5.d)

Initial	Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)	
GG	3	Malfunctions that occur after EOP entry: 1-2
GG	2	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	2	EOPs entered/requiring substantive actions: 1-2
GG	0	EOP contingencies requiring substantive actions: 0-2
GG	2	Crew Critical Tasks: 2-3
GG		

COMMENTS:

ATTACHMENT 5
ESG CRITICAL TASKS

14-01 ILOT NRC ESG-2

CT#1 (Salem Critical Operator Action) Terminate an Inadvertent SI by stopping all but 21 or 22 charging pump within 23 minutes of SI initiation.

Bases: Failing to terminate an Inadvertent SI leads to PZR overfill and water relief through the PZR PORVs/Safeties (Engineering Eval S-2-RC-MEE-1108)

CT#2 (Salem Critical Operator Action) Terminate an Inadvertent SI by re-establishing Letdown within 45 minutes of the inadvertent SI.

Bases: Failing to terminate an Inadvertent SI leads to PZR overfill and water relief through the PZR PORVs/Safeties (Engineering Eval S-2-RC-MEE-1108)

EVENTS LEADING TO CORE DAMAGE

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
N	TRANSIENTS with PCS Unavailable	N	Loss of Service Water
N	Steam Generator Tube Rupture	N	Loss of CCW
N	Loss of Offsite Power	N	Loss of Control Air
N	Loss of Switchgear and Pen Area Ventilation	N	Station Black Out
N	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
N	Containment Sump Strainers	N	Gas Turbine
N	SSWS Valves to Turbine Generator Area	N	Any Diesel Generator
N	RHR Suction Line valves from Hot Leg	N	Auxiliary Feed Pump
N	CVCS Letdown line Control and Isolation Valves	N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
N	Cooldown the RCS and depressurize the system
N	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
N	Initiate feed and bleed

Complete this evaluation form for each ESG.

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